Linear Algebra Cheatsheet

Guide to Matrices, Subspaces, and Factorizations

By Roberto León Landeros

1. Vectors, Matrices & Spaces Fundamentals

Vector Space: A set V closed under addition and scalar multiplication satisfying 8 axioms: commutativity ($\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$), associativity ($(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$), zero vector ($\mathbf{v}+\mathbf{0}=\mathbf{v}$), additive inverse ($\mathbf{v}+(-\mathbf{v})=\mathbf{0}$), multiplicative identity ($1\mathbf{v}=\mathbf{v}$), distributivity of scalar sums ($(c+d)\mathbf{v}=c\mathbf{v}+d\mathbf{v}$), distributivity of vector sums ($c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$), and scalar associativity ($c(d\mathbf{v})=(cd)\mathbf{v}$).
Basic Operations: For $\mathbf{u}=(u_1,u_2,\ldots,u_n)$ and $\mathbf{v}=(v_1,v_2,\ldots,v_n)$:

- Addition: $\mathbf{u}+\mathbf{v}=(u_1+v_1,u_2+v_2,\ldots,u_n+v_n)$
- Scalar Multiplication: $c\mathbf{u}=(cu_1,cu_2,\ldots,cu_n)$
- Cross Product ($\mathbb{R}^3$): $\mathbf{u}\times\mathbf{v}=\det\begin{bmatrix}\hat{\mathbf{i}}&\hat{\mathbf{j}}&\hat{\mathbf{k}}\\u_1&u_2&u_3\\v_1&v_2&v_3\end{bmatrix}$
  $=(u_2v_3-u_3v_2)\hat{\mathbf{i}}-(u_1v_3-u_3v_1)\hat{\mathbf{j}}+(u_1v_2-u_2v_1)\hat{\mathbf{k}}$
- Norm (Length): $\|\mathbf{u}\|=\sqrt{\mathbf{u}\cdot\mathbf{u}}$

Geometry & Inequalities: Cauchy-Schwarz: $|\mathbf{u}\cdot\mathbf{v}|\le\|\mathbf{u}\|\|\mathbf{v}\|$. Triangle Inequality: $\|\mathbf{u}+\mathbf{v}\|\le\|\mathbf{u}\|+\|\mathbf{v}\|$. Angle between vectors: $\cos\theta=\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$.
Dependence & Basis: Linearly Independent if $c_1\mathbf{v}_1+\cdots+c_n\mathbf{v}_n=0\implies c_i=0$ for all i. A basis is a linearly independent set that spans the space.
For subspaces A and B, the dimension formula is $\dim(A+B)=\dim(A)+\dim(B)-\dim(A\cap B)$.
Let $\mathbf{u},\mathbf{v}\in\mathbb{R}^n$. The dot product is $\mathbf{u}\cdot\mathbf{v}=\mathbf{u}^T\mathbf{v}=\sum u_iv_i=\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$.
Orthogonality: $\mathbf{u}\perp\mathbf{v}\iff\mathbf{u}\cdot\mathbf{v}=0$. A matrix $A\in\mathbb{R}^{m\times n}$ represents a linear transformation $T:\mathbb{R}^n\to\mathbb{R}^m$.
The identity matrix I acts as the multiplicative unit ($AI=IA=A$); $I=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$.

2. Views of Matrix Multiplication

Let $A\in\mathbb{R}^{m\times n}$ and $B\in\mathbb{R}^{n\times p}$. The product $C=AB$ is an $m\times p$ matrix.

- **1. Standard (Dot Product):** $c_{ij}=\sum_{k=1}^{n}a_{ik}b_{kj}=(\text{row } i \text{ of } A)\cdot(\text{col } j \text{ of } B)$.
- **2. Column View (Linear Combination of Columns):** The columns of C are combinations of the columns of A.

$$AB=A\begin{bmatrix}|&&|\\\mathbf{b}_1&\cdots&\mathbf{b}_p\\|&&|\end{bmatrix}=\begin{bmatrix}|&&|\\A\mathbf{b}_1&\cdots&A\mathbf{b}_p\\|&&|\end{bmatrix}$$

- **3. Row View (Linear Combination of Rows):** The rows of C are combinations of the rows of B.

$$AB=\begin{bmatrix}-&\mathbf{a}_1&-\\&\vdots&\\-&\mathbf{a}_m&-\end{bmatrix}B=\begin{bmatrix}-&\mathbf{a}_1B&-\\&\vdots&\\-&\mathbf{a}_mB&-\end{bmatrix}$$

- **4. Outer Product (Sum of Rank-1 Matrices):** AB is the sum of columns of A times rows of B.

$$AB=\sum_{k=1}^{n}(\text{col } k \text{ of } A)(\text{row } k \text{ of } B)=\sum_{k=1}^{n}\mathbf{a}_k\mathbf{b}_k^T$$

- **5. Block Matrices (Submatrices):** Matrices can be multiplied as scalars if blocks are compatible.

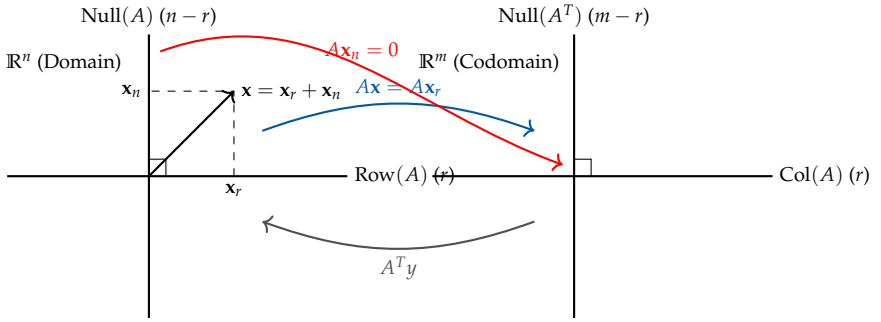
$$\begin{bmatrix}A&B\\C&D\end{bmatrix}\begin{bmatrix}E&F\\G&H\end{bmatrix}=\begin{bmatrix}AE+BG&AF+BH\\CE+DG&CF+DH\end{bmatrix}$$

3. The Four Fundamental Subspaces

Let $A\in\mathbb{R}^{m\times n}$ have rank r (number of pivots). The Fundamental Theorem of Linear Algebra states:

Subspace	Dim	Basis & Location
$\text{Col}(A)$ (Column Space)	r	Pivot columns of A. Lives in $\mathbb{R}^m$.
$\text{Null}(A)$ (Nullspace)	$n-r$	Special solutions to $Ax=0$. Lives in $\mathbb{R}^n$.
$\text{Row}(A)=\text{Col}(A^T)$	r	Non-zero rows of RREF. Lives in $\mathbb{R}^n$.
$\text{Null}(A^T)$ (Left Nullspace)	$m-r$	Solutions to $A^Ty=0$. Lives in $\mathbb{R}^m$.

Orthogonality: $\text{Row}(A)\perp\text{Null}(A)$ in $\mathbb{R}^n$ and $\text{Col}(A)\perp\text{Null}(A^T)$ in $\mathbb{R}^m$. Any $x\in\mathbb{R}^n$ can be split into $x=x_r+x_n$.



4. Systems of Equations & Elimination

To solve $Ax = b$, we construct $[A \mid b]$ and apply row operations to reach Reduced Row Echelon Form (RREF).

- **Elementary Matrices (E):** Left-multiply to perform row operations. E.g., subtract l_{21} times row 1 from row 2:

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{21}A = \begin{bmatrix} \text{row } 1 \\ \text{row } 2 - l_{21}(\text{row } 1) \\ \text{row } 3 \end{bmatrix}$$

- **Permutation Matrices (P):** Reorder rows (left-multiply) or columns (right-multiply). $P^{-1} = P^T$.
- **Existence & Uniqueness:** $Ax = b$ has a solution $\iff b \in \text{Col}(A)$. Unique solution $\iff \text{Null}(A) = \{0\}$ (full column rank, $r = n$).
- **Left/Right Inverses:** If $r = n$ (full column rank), A has a Left Inverse: $A_{left}^{-1} = (A^TA)^{-1}A^T$. If $r = m$ (full row rank), A has a Right Inverse: $A_{right}^{-1} = A^T(AA^T)^{-1}$.
- **Rouché-Frobenius Theorem:** No solution if $\text{rank}(A) < \text{rank}([A \mid b])$. Unique solution if $\text{rank}(A) = \text{rank}([A \mid b]) = n$. Infinite solutions if $\text{rank}(A) = \text{rank}([A \mid b]) < n$.

5. Determinants & Inverses

The determinant $\det(A)$ or $|A|$ is only defined for square $n \times n$ matrices. Geometrically, it measures the scaling factor of the linear transformation (Area in $\mathbb{R}^2$, Volume in $\mathbb{R}^3$ of the parallelepiped formed by the columns).

- **Properties:**

$$\begin{aligned} \det(AB) &= \det(A)\det(B) \\ \det(A^T) &= \det(A) \\ \det(A^{-1}) &= 1/\det(A) \\ \det(cA) &= c^n \det(A) \\ \det(A+B) &\neq \det(A) + \det(B) \text{ (generally)} \end{aligned}$$

- Row exchange flips sign. Multiplying a row by c scales det by c. Adding a multiple of one row to another does not change the determinant.
- Determinant of a triangular matrix is the product of its diagonal entries.
- $\det(A) \neq 0 \iff A$ is invertible $\iff \text{Null}(A) = \{0\}$.
- **Laplace Expansion (Cofactors):** $\det(A) = \sum_{j=1}^{n} a_{ij}C_{ij}$, where $C_{ij} = (-1)^{i+j}\det(M_{ij})$.
- **Leibniz Formula:** $\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}$, where σ are permutations and sgn is the parity.
- **Practical Inverse (Adjugate):** $A^{-1} = \frac{1}{\det(A)}C^T$.
- **Block Matrices Inverse:** If A, D are invertible: $\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & 0 \\ 0 & D^{-1} \end{bmatrix}$
- **Cramer's Rule:** To solve $Ax = b$, $x_i = \frac{\det(A_i)}{\det(A)}$, where A_i is A with its i-th column replaced by b.

6. Factorizations: The Big Picture

Decomposing a matrix reveals its fundamental structure and simplifies complex operations.

A. LU Decomposition (Elimination)

$A = LU$. Separates A into a lower triangular L (multipliers) and upper triangular U (pivots). If row swaps are needed, $PA = LU$.

B. QR Decomposition (Gram-Schmidt)

$A = QR$. Q has orthonormal columns ($Q^TQ = I$), R is upper triangular. Used for stable least squares and eigenvalue algorithms. Iterative orthogonal projection: $\mathbf{u}_k = \mathbf{a}_k - \sum_{i=1}^{k-1} \text{proj}_{\mathbf{u}_i}(\mathbf{a}_k)$.

C. Eigen-Decomposition (Diagonalization)

$A = X\Lambda X^{-1}$. X contains eigenvectors as columns, Λ has eigenvalues on the diagonal. Requires n independent eigenvectors.

$$A = \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \dots & \mathbf{x}_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} X^{-1}$$

D. Spectral Theorem (Symmetric Matrices)

If $A = A^T$, eigenvalues are real, and eigenvectors are orthogonal. $A = Q\Lambda Q^T$, where Q is an orthogonal matrix.

E. Singular Value Decomposition (SVD)

The crown jewel. Exists for ANY $m \times n$ matrix. $A = U\Sigma V^T$.

- $U \in \mathbb{R}^{m \times m}$ (Orthogonal, left singular vectors, basis for $\text{Col}(A), \text{Null}(A^T)$).
- $V \in \mathbb{R}^{n \times n}$ (Orthogonal, right singular vectors, basis for $\text{Row}(A), \text{Null}(A)$).
- $\Sigma \in \mathbb{R}^{m \times n}$ (Diagonal, singular values $\sigma_i = \sqrt{\lambda_i(A^TA)}$).
- **Pseudoinverse:** The Moore-Penrose pseudoinverse is $A^+ = V\Sigma^+U^T$.

7. Eigenvalues & Eigenvectors

$\mathbf{x}$ is an eigenvector of A if $A\mathbf{x} = \lambda\mathbf{x}$. The action of the matrix simplifies to a scalar multiplication.

- **Characteristic Equation:** $\det(A - \lambda I) = 0$. Roots are λ_i.
- **Nullspace Link:** Eigenvectors are the basis of $\text{Null}(A - \lambda I)$.
- **Multiplicity & Diagonalization:** A is diagonalizable $\iff$ algebraic multiplicity equals geometric multiplicity ($\dim(\text{Null}(A - \lambda I))$) for all eigenvalues.
- **Trace & Det:** $\sum \lambda_i = \text{tr}(A)$ and $\prod \lambda_i = \det(A)$.
- **Powers:** $A^k = X\Lambda^k X^{-1}$. $A^k\mathbf{x} \to 0$ if all $|\lambda_i| < 1$.
- **Markov Matrices:** Columns sum to 1. Always has $\lambda = 1$. The corresponding eigenvector is the steady state.

8. Orthogonality & Projections

A matrix Q with orthonormal columns satisfies $Q^TQ = I$. If Q is square, $Q^T = Q^{-1}$. Multiplication by Q preserves lengths: $\|Q\mathbf{x}\| = \|\mathbf{x}\|$.

Projections

To project a vector $\mathbf{b}$ onto $\text{Col}(A)$, the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ must be orthogonal to $\text{Col}(A)$, meaning $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$.

- **Normal Equations:** $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$.
- **Projection Matrix:** $P = A(A^TA)^{-1}A^T$. Properties: $P^2 = P$ and $P^T = P$.
- **1D Projection:** Onto a line $\mathbf{a}$: $P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}$.

9. Linear Transformations

A linear transformation $T(\mathbf{x}) = A\mathbf{x}$ obeys $T(c\mathbf{x} + \mathbf{y}) = cT(\mathbf{x}) + T(\mathbf{y})$.

- **Matrix Representation:** The j-th column of A is $T(\mathbf{e}_j)$.
- **Rotation:** Counterclockwise by θ: $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
- **Change of Basis & Similar Matrices:** If B is the matrix of basis vectors, a vector $\mathbf{v}$ in standard basis relates to coordinates $\mathbf{c}$ in B-basis by $\mathbf{v} = B\mathbf{c}$. Similar matrices represent the same transformation in different bases: $M = B^{-1}AB$. They share the same eigenvalues, trace, and determinant.
- **Rank-Nullity Theorem:** For a linear transformation $T : V \to W$, $\dim(V) = \text{rank}(T) + \text{nullity}(T)$.

10. Special Matrices Summary

- **Symmetric:** $A = A^T$. Real eigenvalues, orthogonal eigenvectors.
- **Positive Definite:** $A = A^T$ and $\mathbf{x}^TA\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$. All $\lambda_i > 0$. All pivots > 0. $A = R^TR$ (Cholesky). **Sylvester's Criterion:** All leading principal minors have positive determinants ($\det > 0$).
- **Orthogonal:** $Q^TQ = I$. Columns form an orthonormal basis. $\det(Q) = \pm 1$. Eigenvalues $|\lambda| = 1$.
- **Rank-1:** $A = \mathbf{u}\mathbf{v}^T$. $\text{Col}(A)$ is the line through $\mathbf{u}$. $\text{Row}(A)$ is the line through $\mathbf{v}$. Has $\lambda = \mathbf{v}^T\mathbf{u}$ and $n - 1$ zero eigenvalues.
- **Skew-Symmetric:** $A = -A^T$. Pure imaginary eigenvalues.
- **Idempotent:** $P^2 = P$. Typical of projection matrices.
- **Nilpotent:** $A^k = 0$ for some integer k.

11. Quadratic Forms & Matrix Exponentials

- **Quadratic Forms:** Expressed as $f(\mathbf{x}) = \mathbf{x}^TA\mathbf{x}$ for a symmetric matrix A. The Hessian matrix contains second partial derivatives, critical for multivariable optimization.
- **Matrix Exponential (e^{At}):** $e^{At} = I + At + \frac{1}{2!}(At)^2 + \cdots = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}$. Used to solve linear systems of differential equations $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$, with solution $\mathbf{u}(t) = e^{At}\mathbf{u}(0)$.

12. Mini-Algorithms (Pipelines)

- **Solve $Ax = b$:** $[A \mid b] \xrightarrow{\text{RREF}}$ Check pivots: $\begin{cases} \text{Row } [0 \dots 0 \mid c \neq 0] \implies \textbf{No Solution} \\ r = n \implies \textbf{Unique} \\ r < n \implies \textbf{∞ Solutions (Free Vars)} \end{cases}$
- **Subspace Bases:** $A \xrightarrow{\text{RREF}} \implies \begin{cases} \text{Col}(A) : \text{Original cols of } A \text{ at pivot positions} \\ \text{Null}(A) : \text{Set one free var} = 1 \text{ (others 0)} \to \text{Solve} \end{cases}$
- **Rank (r):** $A \xrightarrow{\text{Echelon Form}} \implies r = \text{Total number of pivots}$
- **Eigenvalues & Vectors:** $\det(A - \lambda I) = 0 \xrightarrow{\text{Roots}} \lambda_i \implies$ Find basis for $\text{Null}(A - \lambda_i I) \xrightarrow{\text{Yields}} \mathbf{x}_i$
- **Diagonalization ($A = X\Lambda X^{-1}$):** Find n independent $\mathbf{x}_i \implies X = [\mathbf{x}_1 \dots \mathbf{x}_n] \implies \Lambda = \text{diag}(\lambda_1 \dots \lambda_n)$
- **Gram-Schmidt:** $\mathbf{u}_1 = \mathbf{a}_1 \implies \mathbf{u}_k = \mathbf{a}_k - \sum \text{proj}_{\mathbf{u}_j}(\mathbf{a}_k) \implies$ Normalize: $\mathbf{q}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$
- **Least Squares ($\hat{x}$):** $Ax = b$ (inconsistent) $\xrightarrow{\text{Left-mult } A^T} A^T A\hat{\mathbf{x}} = A^T\mathbf{b} \implies$ Solve for $\hat{\mathbf{x}}$

13. The Big Picture (Concept Graphs)

- **LU** $\implies$ Gaussian Elimination $\implies$ Solves $Ax = b$ efficiently for multiple $\mathbf{b}$'s
- **QR** $\implies$ Orthogonality $\implies$ Projections $\implies$ Stable Least Squares ($R\hat{\mathbf{x}} = Q^T\mathbf{b}$)
- **SVD** $\implies$ Rotations + Scaling ($U\Sigma V^T$) $\implies$ Full Structure $\implies$ Defines 4 Fundamental Subspaces
- **Eigen** $\implies$ Diagonalization ($A = X\Lambda X^{-1}$) $\implies$ Decouples Variables $\implies$ Dynamics & Powers (A^k)

14. Common Pitfalls ($\neq$)

- $\mathbf{AB} \neq \mathbf{BA} \implies$ Matrix multiplication is **not commutative**. Order is strict.
- $\det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B}) \implies$ Determinant is entirely non-linear over addition.
- **Independence $\not\Rightarrow$ Orthogonality** $\implies$ Orthogonal vectors are independent, but not vice versa.
- **Invertible $\not\Leftrightarrow$ Diagonalizable** $\implies$ Totally unrelated properties (e.g., $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ invertible but not diag).
- **Order Reversal** $\implies$ Transposes & inverses flip order: $(AB)^T = B^T A^T$ and $(AB)^{-1} = B^{-1}A^{-1}$.
- **Projection Trap** $\implies$ For $P = A(A^TA)^{-1}A^T$, you **cannot** distribute the inverse unless A is square.

15. Key Intuitions ($\to$)

- **Determinant** $\to$ Volume scaling factor. ($\det = 0 \implies$ space collapses, information is lost).
- **Eigenvalues** $\to$ Stretch/shrink factors along invariant directions.
- **Rank** $\to$ True dimensionality (information content) of the output space.
- **Nullspace** $\to$ Invisibility. Inputs that get crushed to the zero vector.
- **Projections** $\to$ Closest "shadow" in a subspace. Error vector is strictly $\perp$ to the subspace.

16. Exam Tips & Verifications

- **Check Eigenvalues:** Always verify $\sum \lambda_i = \text{tr}(A)$ (diagonal sum) and $\prod \lambda_i = \det(A)$.
- **Symmetric Shortcut:** $A = A^T \implies$ Always diagonalizable, real λ's, orthogonal eigenvectors.
- **Orthogonal Shortcut:** $Q^TQ = I \implies Q^{-1} = Q^T$ (Zero calculation required for inverse).
- **Zero Eigenvalue:** $\lambda = 0 \implies \det(A) = 0 \implies$ Singular $\implies \text{Null}(A)$ contains non-zero vectors.
- **Dependence Spotting:** Proportional rows/cols or all-zero row $\implies \det = 0$ and $r < n$.